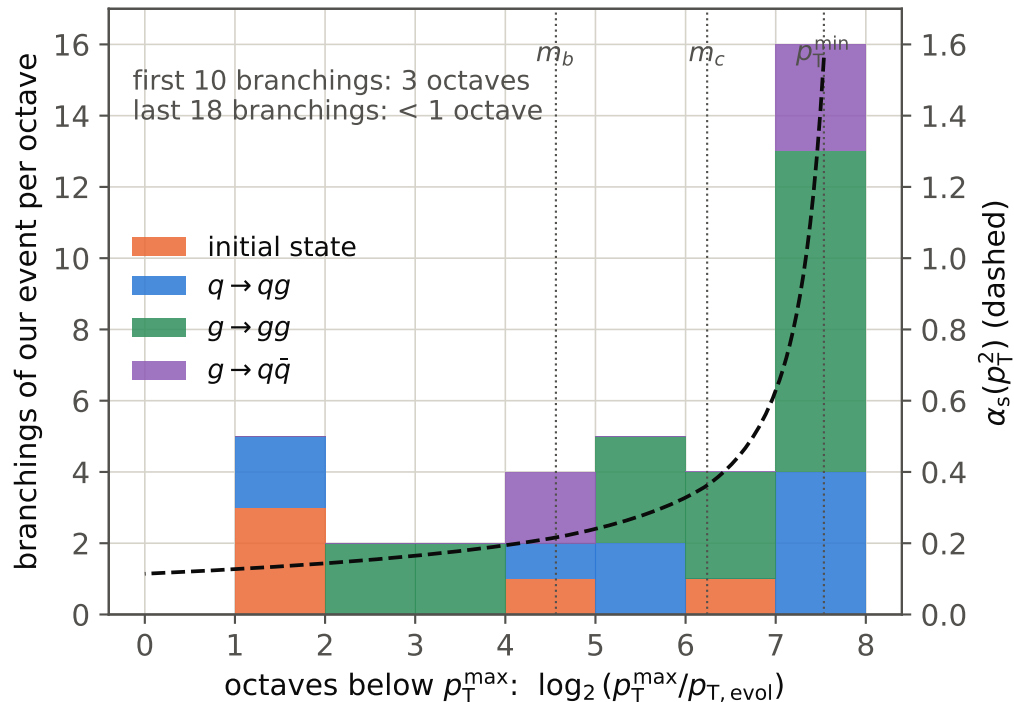
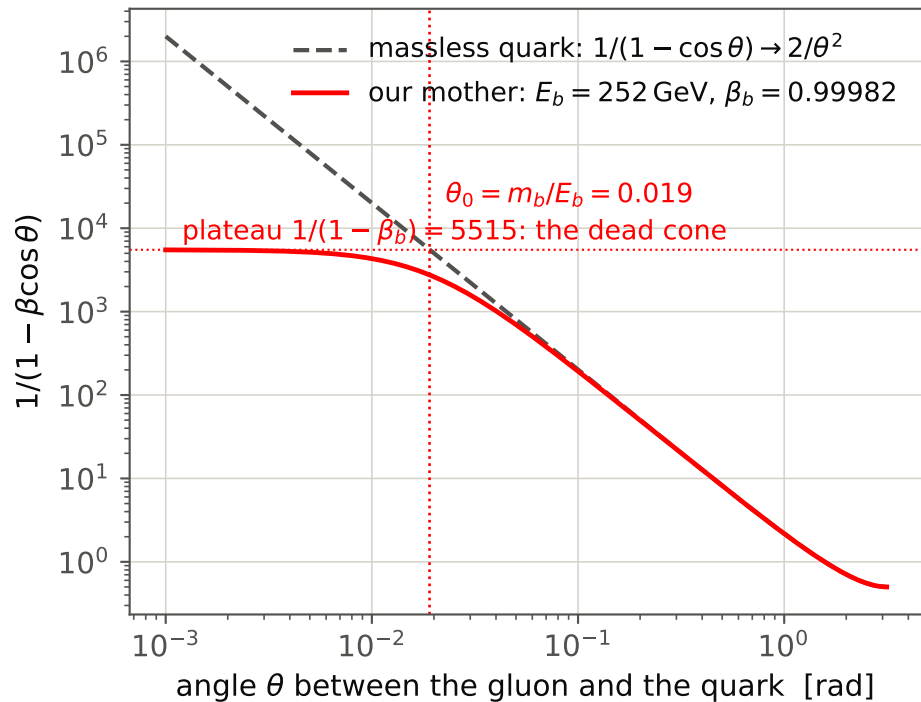


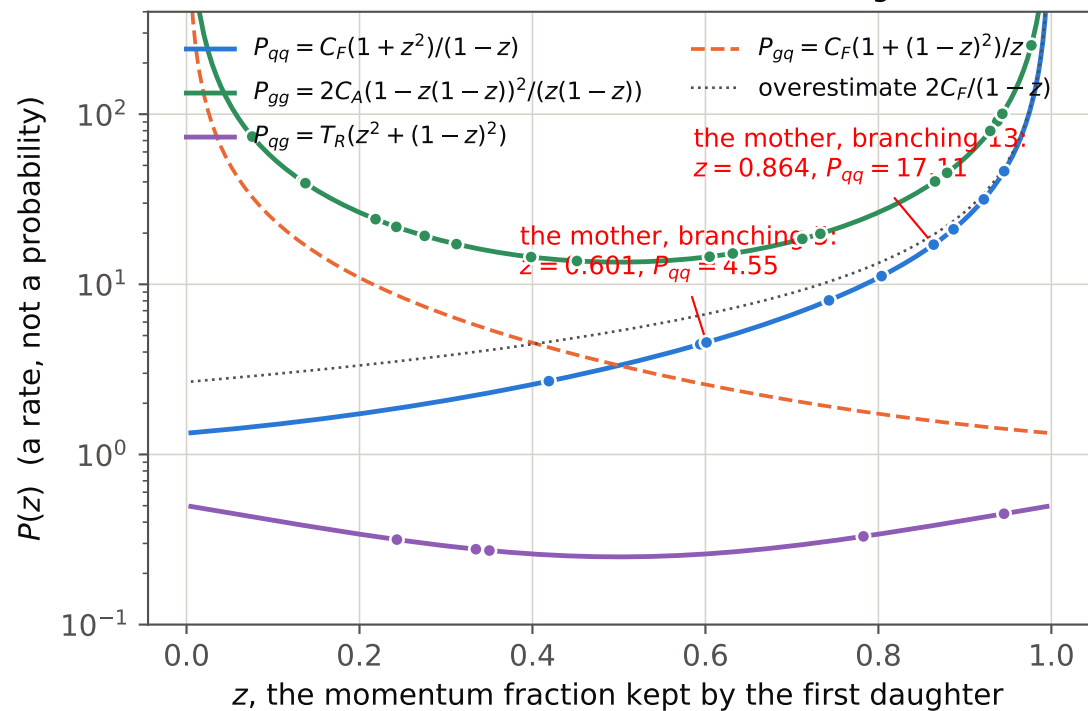
Why a quark cannot travel alone: the propagator $1/(1 - \beta \cos \theta)$ and the branchings per octave of our event

(a) the collinear singularity, and where the mass stops it (b) 38 branchings: the rate per octave grows as the scale falls

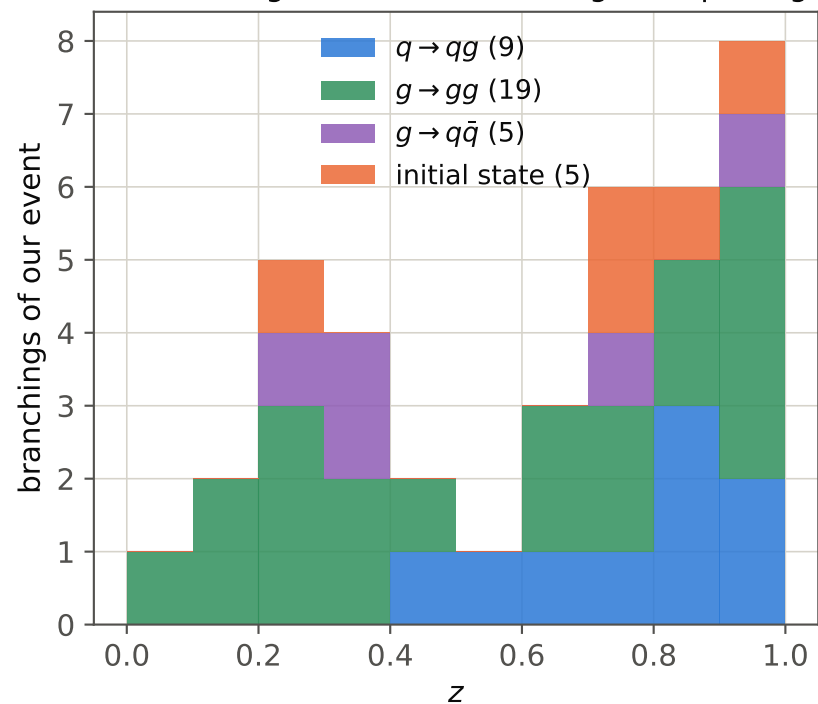


The four splitting functions, and the z values the shower of our event actually drew

(a) the kernels; dots: the 33 final-state branchings of our event

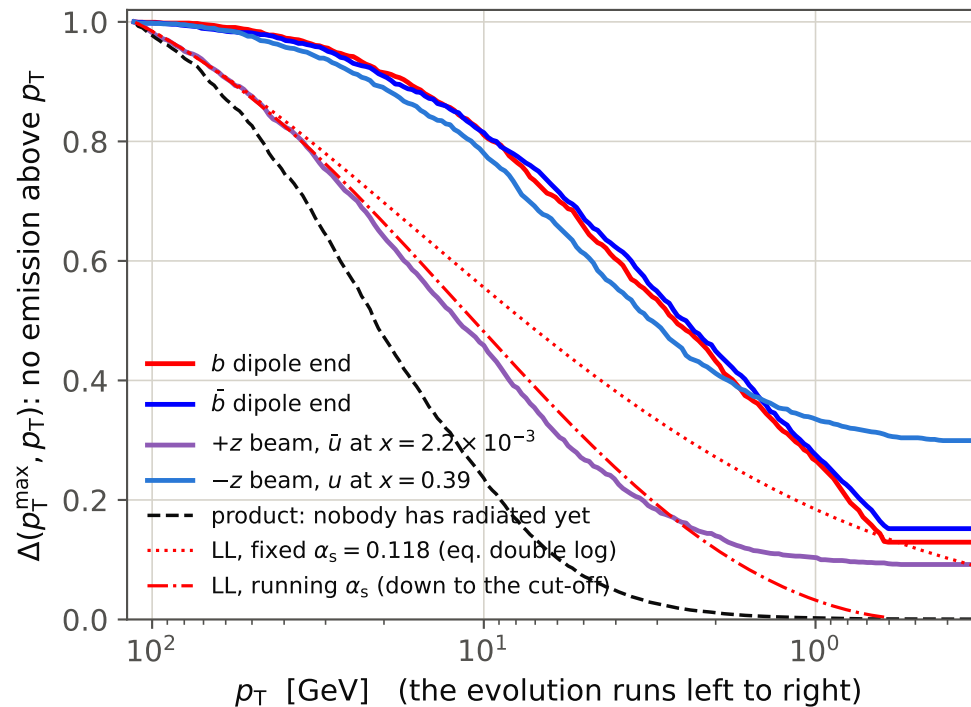


(b) soft daughters dominate the gluon splittings

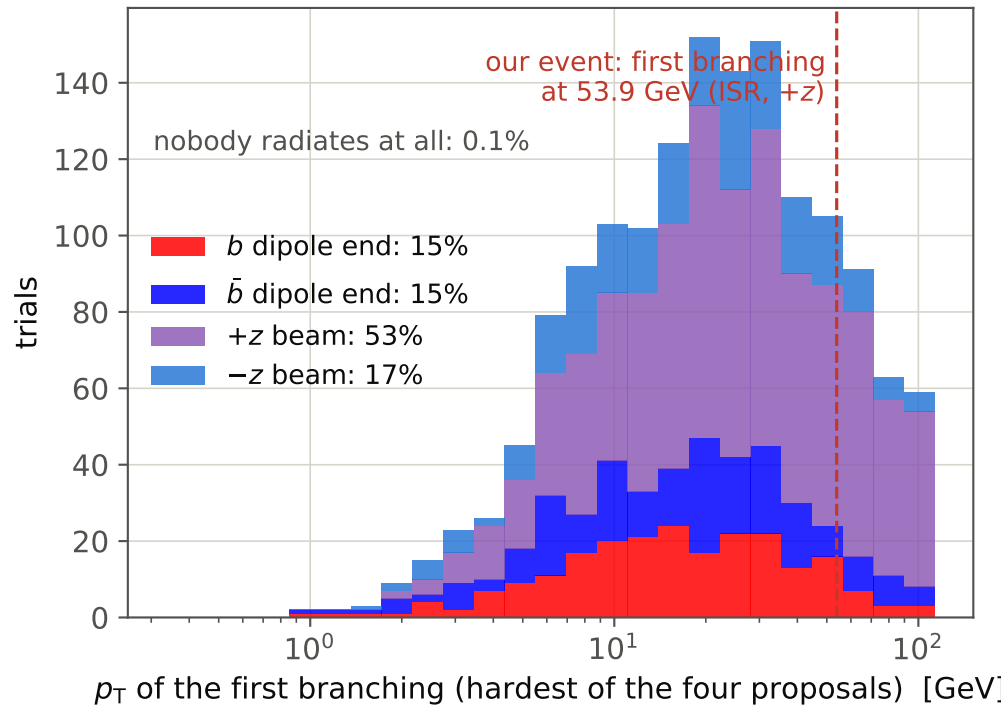


The Sudakov form factor: measured from 1500 proposals per radiator, against the leading-log formula

(a) at 10 GeV: measured 0.81, LL fixed 0.56, LL running 0.48

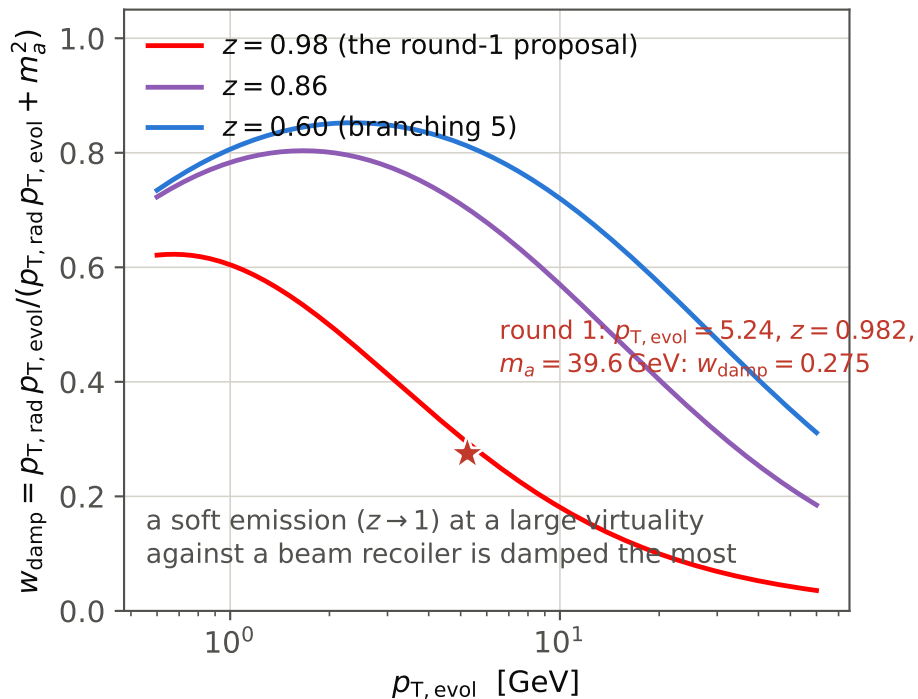
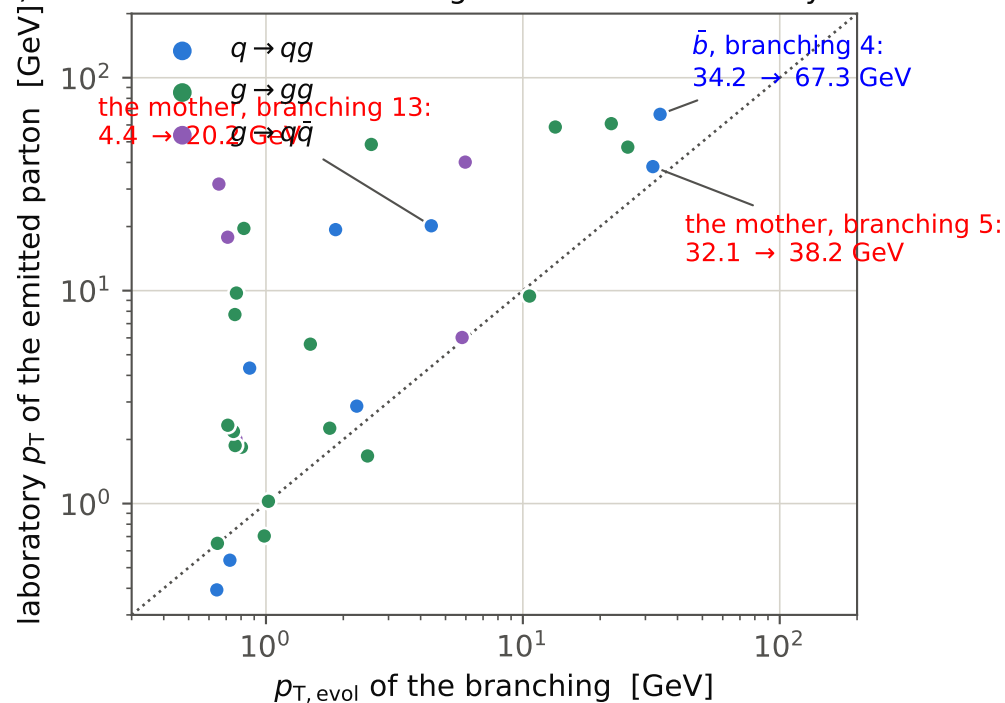


(b) who radiates first: the +z beam, at small x , most often



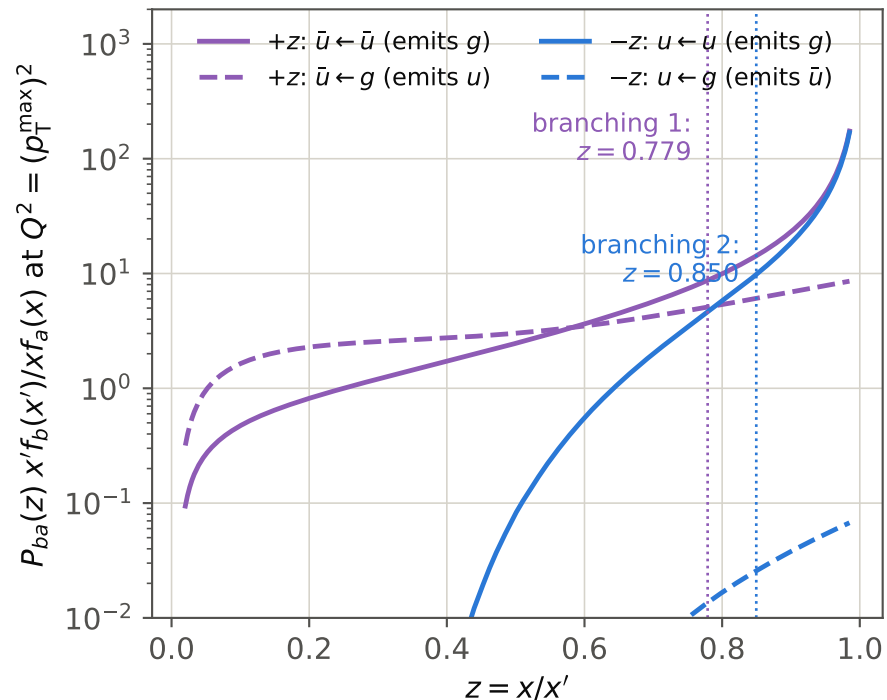
Final-state radiation: the evolution variable is not the gluon's laboratory p_T , and the beam-recoil damping

(a) the 33 final-state branchings: a factor 3 either way is common (b) TimeShower:dampenBeamRecoil for a radiator of $p_T = 113$ GeV

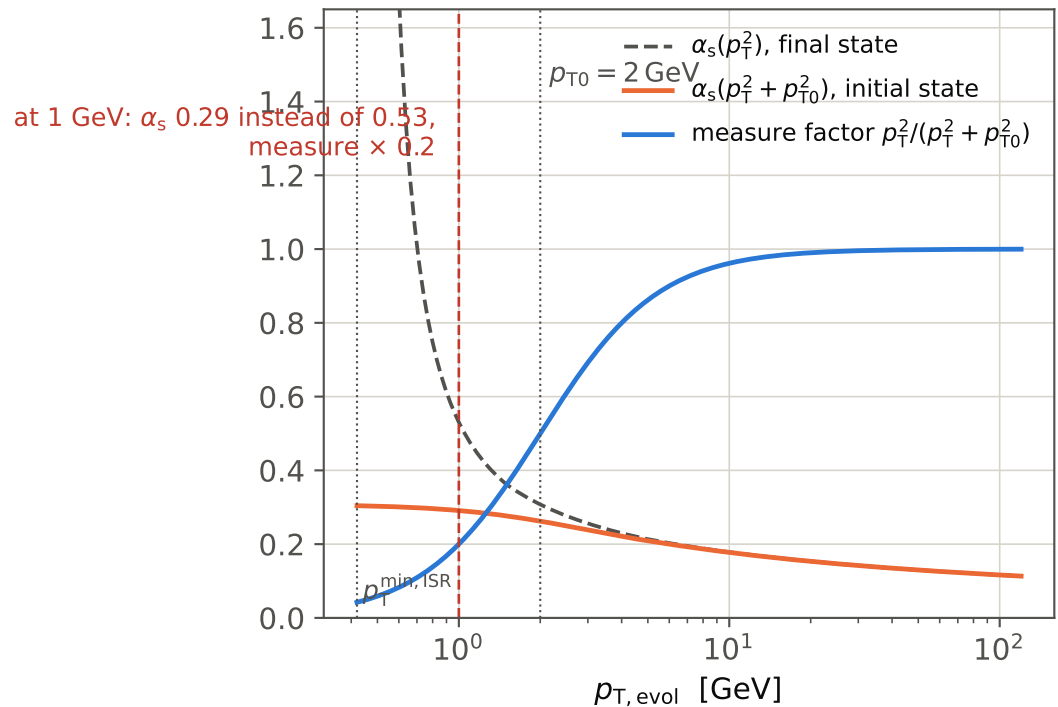


Backward evolution: the kernel times the PDF ratio for the two beams of our event, and the infrared regularisation

(a) small x : $g \rightarrow q\bar{q}$; a valence quark: $q \rightarrow qg$

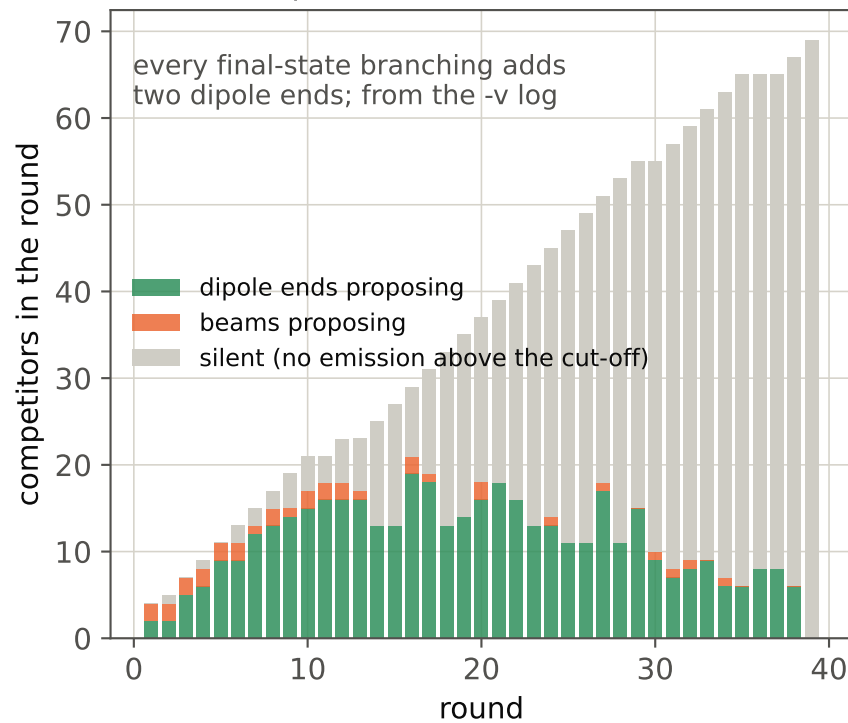
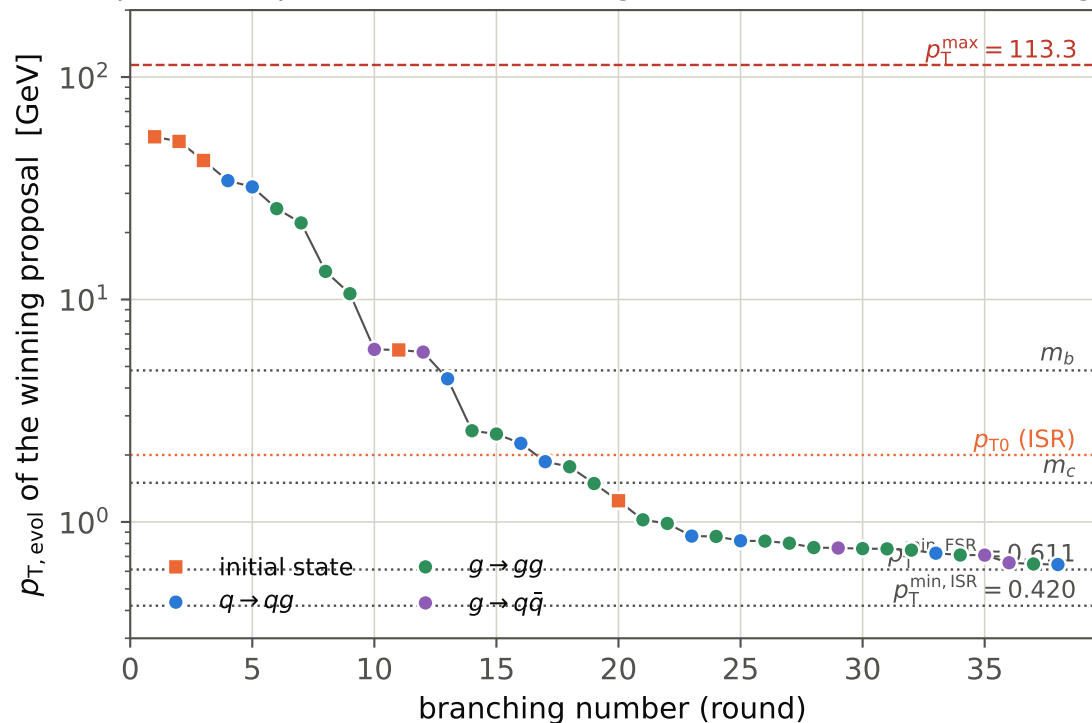


(b) pT0Ref tames coupling and measure near the cut-



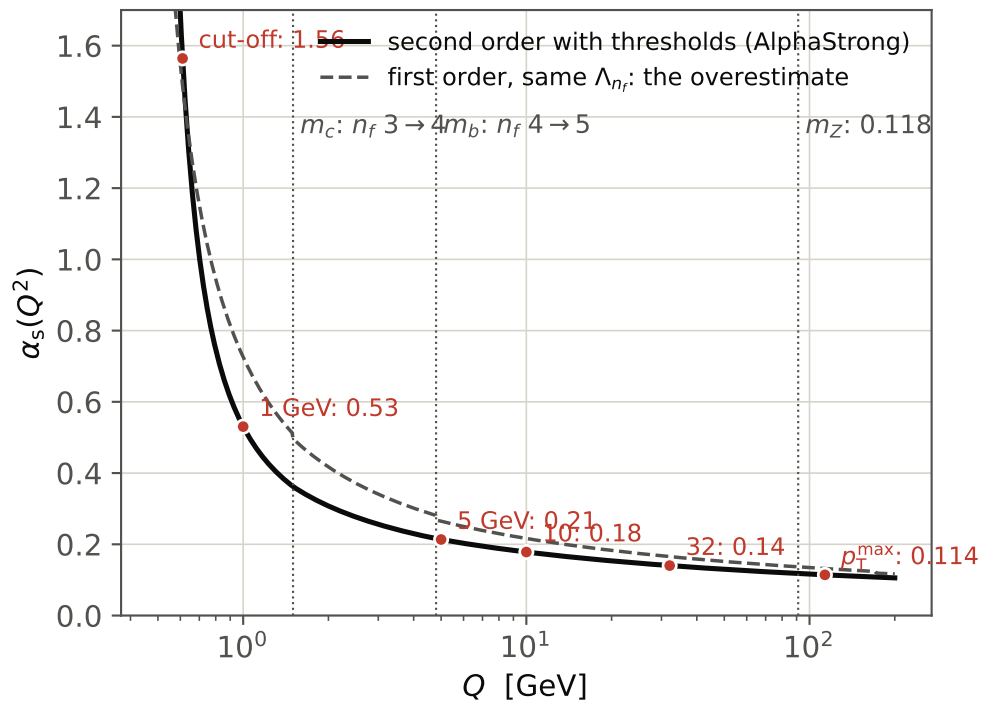
One ordered sequence: the ladder of 38 rungs of our event and the competitors of every round

(a) steep at the top (3 octaves in 10 rungs), flat at the bottom (18 rungs in one) (b) 4 competitors in round 1, 69 in round 39

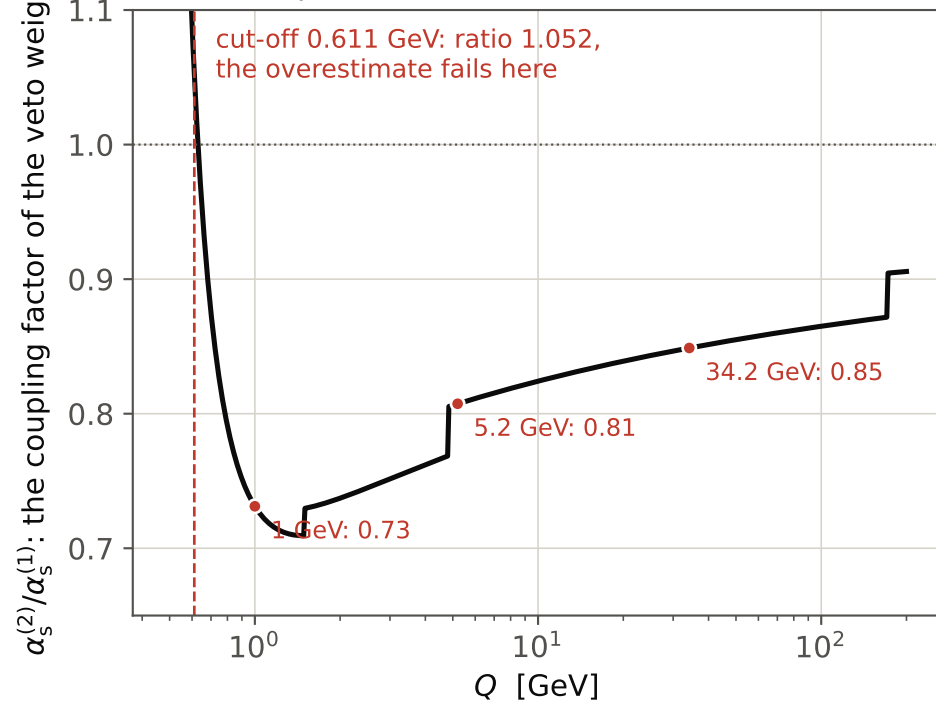


The CP5 shower coupling: $\alpha_s(m_Z) = 0.118$ at second order with thresholds, and its first-order overestimate

(a) $\Lambda_5 = 0.226, \Lambda_4 = 0.328, \Lambda_3 = 0.382$ GeV

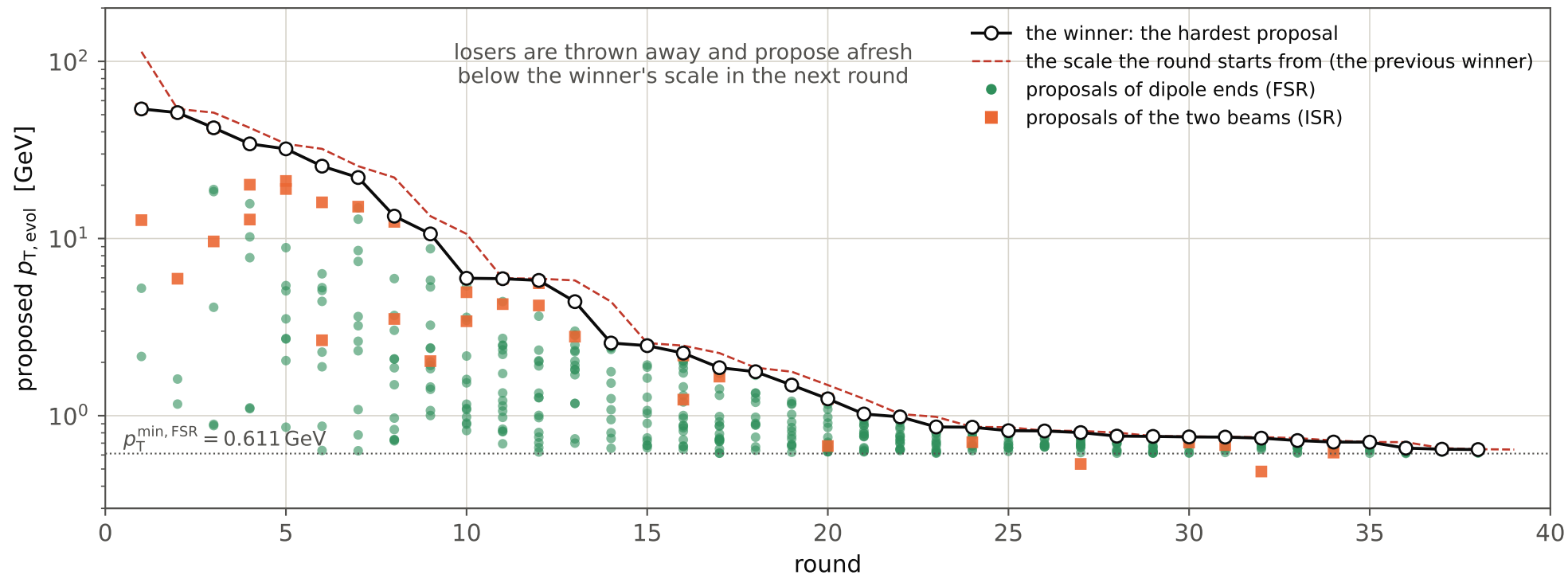


(b) the price of the overestimate: 0.73 to 0.87

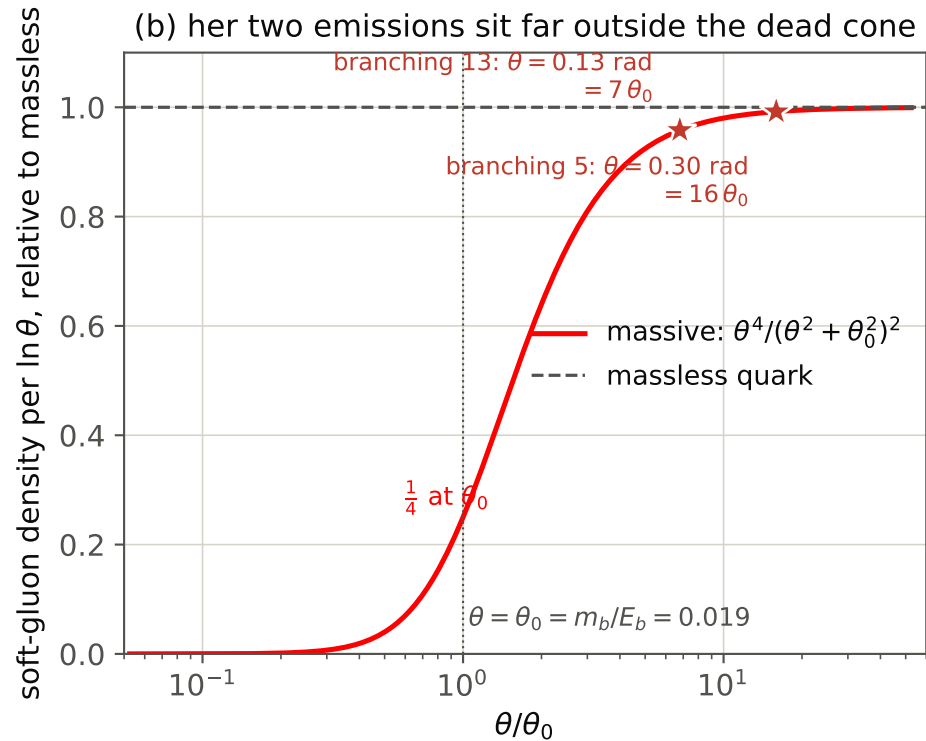
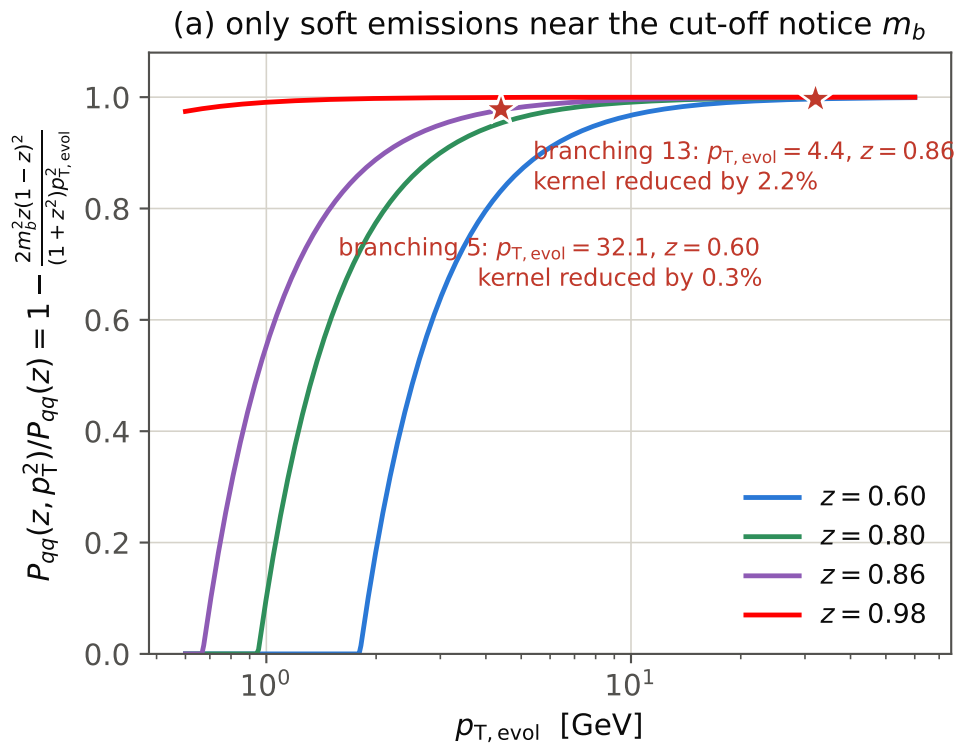


Everybody proposes, the hardest wins: every proposal of every round of our event, from the -v log

round 1: 2.2, 5.2 (FSR), 53.9, 12.7 (ISR) GeV; 39 rounds, 467 proposals in all, 38 winners

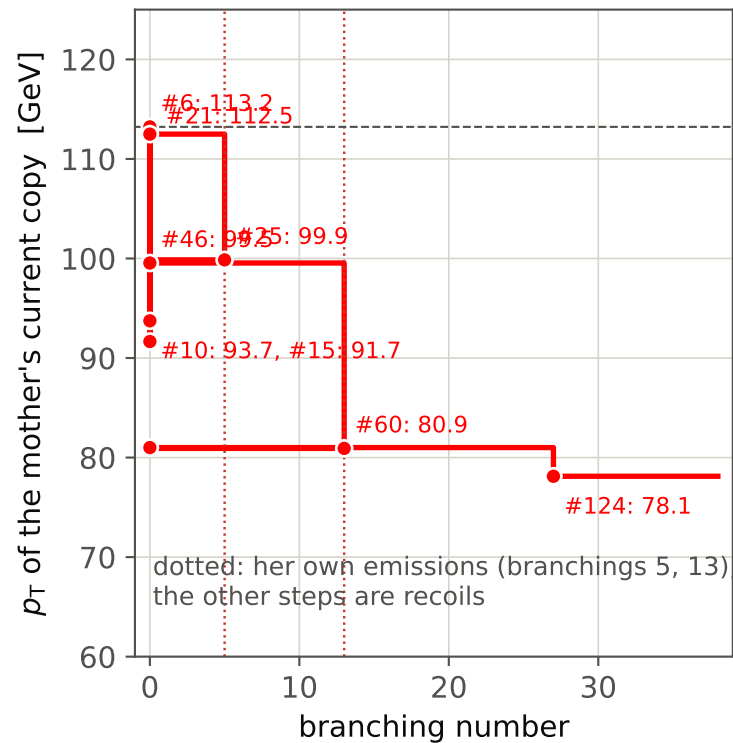


The dead cone of the heavy mother: the mass term of the $b \rightarrow bg$ kernel and the angular density of soft gluons

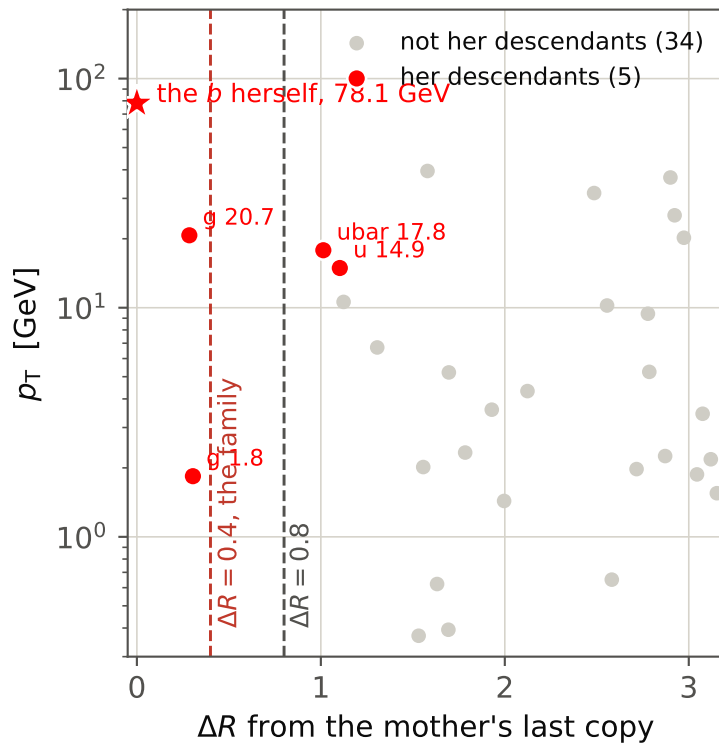


Children who run away: the mother's p_T copy by copy, her descendants, and the family inside a cone of radius R

(a) 113.2 $\rightarrow$ 93.7 $\rightarrow$ 91.7 $\rightarrow$ 112.5 $\rightarrow$ 99.9 $\rightarrow$ 80.9 $\rightarrow$ 78.1 GeV



(b) her descendants: the $u\bar{u}$ pair at $\Delta R \approx 1$



(c) the ledger against the cone radius

